

quadratic equations

& STARTER PLAN

Generated 14 July 2026 · fundoi.synapex.co.zw

Ø=ÜÜ Quadratic Equations – Study Notes

Level: O Level (Form /6) – General Mathematics

Curriculum: ZIMSEC & Cambridge (IGCSE)

1p ã Overview / Introduction

A **quadratic equation** is any algebraic equation that can be written in the form

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

- The highest power of the variable **x** is 2 ! “quadratic”.
- It represents a **parabola** when graphed on the coordinate plane.
- Solving the equation means finding the **roots** (or **zeros**) – the **x** values where the parabola cuts the **x** axis.

Quadratics appear everywhere in O Level maths: motion problems, geometry, optimisation, finance, and in many past paper questions. Mastery of the methods below will give you a solid toolkit for the exam.

2p ã Key Concepts

| Concept | What you need to know | Typical exam use |

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| **Standard form** | $(ax^2 + bx + c = 0)$ (with (a, b, c) integers or simple fractions) | All methods start from this form |

| **Factorisation** | Write the quadratic as $((px + q)(rx + s) = 0)$. Works when the roots are rational integers or simple fractions. | Quick 1 mark questions |

| **Completing the square** | Transform $(ax^2 + bx + c)$ into $(a \bigl(x + \frac{b}{2a}\bigr)^2 = k)$. Useful for deriving the quadratic formula and for solving equations where factorisation is hard. | 2 mark questions, deriving vertex form |

| **Quadratic formula** |

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

| (derived from completing the square) | Universal method; 3 mark questions |

| **Discriminant** $(\Delta = b^2 - 4ac)$ | Determines the nature of the roots:
• $(\Delta > 0)$! two distinct real roots
• $(\Delta = 0)$! one repeated real root
• $(\Delta < 0)$! two complex (non real) roots | Marks are often awarded for stating the nature of the roots |

| **Vertex form** | $(y = a(x - h)^2 + k)$ where $((h, k))$ is the vertex. Obtained by completing the square. | Graph sketching, optimisation |

| **Symmetry & axis of symmetry** | Axis: $(x = -\frac{b}{2a})$. | Useful for sketching and solving geometry problems |

| **Roots & factors relationship** | If roots are (r_1) and (r_2) then $((x-r_1)(x-r_2)=0)$!' $(ax^2+bx+c = a(x-r_1)(x-r_2))$. | Checking work, constructing equations from given roots |
 | **Applications** | – Projectile motion (range, maximum height)
– Area problems (maximising area of a fenced field)
– Finance (compound interest, break even) | Real world problem solving |

3p ã Important Definitions

Term	Definition

Quadratic equation	An equation of the form $(ax^2+bx+c=0)$ with $(a\neq 0)$.
Root / Zero	A value of (x) that satisfies the equation; i.e., makes the left hand side zero.
Factorisation	Expressing a polynomial as a product of lower degree polynomials.
Complete the square	A method of rewriting a quadratic as a perfect square plus a constant.
Discriminant (Δ)	The expression (b^2-4ac) ; decides the nature of the roots.
Vertex	The highest or lowest point of the parabola; coordinates $((h,k))$.
Axis of symmetry	The vertical line $(x = h)$ that bisects the parabola.
Repeated (double) root	When $(\Delta = 0)$; the parabola touches the x axis at a single point.
Complex root	When $(\Delta < 0)$; expressed as $(p\pm qi)$ where $(i^2=-1)$.

4p ã Worked Examples

%_ **Example 1 – Factorisation (ZIMSEC style, 1 mark)**

Solve: $(2x^2+7x+3=0)$

Solution:

1. Look for two numbers whose product = $(2\times 3=6)$ and sum = (7) .
2. The numbers are **6** and **1**.

$(2x^2+6x+x+3=0)$
 $(2x(x+3)+1(x+3)=0)$
 $((2x+1)(x+3)=0)$

Set each factor to zero:

$(2x+1=0 \Rightarrow x=-\frac{1}{2}, x+3=0 \Rightarrow x=-3)$

Answer: $(x=-\frac{1}{2})$ or $(x=-3)$.

%_ **Example 2 – Quadratic Formula (Cambridge, 3 mark)**

Solve: $(3x^2-4x-5=0)$

Solution:

$(a=3, b=-4, c=-5)$

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Discriminant:

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$$\Delta = b^2 - 4ac = (-4)^2 - 4(3)(-5) = 16 + 60 = 76$$

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$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$= \frac{4 \pm \sqrt{76}}{6}$$

$$= \frac{4 \pm 2\sqrt{19}}{6}$$

$$= \frac{2 \pm \sqrt{19}}{3}$$

\]

Answer: $x = \frac{2 + \sqrt{19}}{3}$ or $x = \frac{2 - \sqrt{19}}{3}$.

%_ Example 3 – Completing the Square (2 mark)

Solve: $x^2 + 6x + 5 = 0$

Solution:

1. Move constant to RHS: $x^2 + 6x = -5$.
2. Half the coefficient of x ($6/2 = 3$); square it ($3^2 = 9$).
3. Add 9 to both sides:

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$$x^2 + 6x + 9 = -5 + 9$$

$$(x+3)^2 = 4$$

\]

4. Take square root: $x+3 = \pm 2$.

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$$x = -3 \pm 2 \Rightarrow x = -1 \text{ or } x = -5$$

\]

Answer: $x = -1$ or $x = -5$.

%_ Example 4 – Application – Maximum Area (4 mark)

A rectangular garden is to be fenced on three sides (the fourth side is a river). The total length of fencing available is **60 m**. Find the dimensions that give the **maximum area**.

Solution:

1. Let the side parallel to the river be x (no fence needed). The two equal sides perpendicular to the river are each y .
2. Fence used: $x + 2y = 60 \Rightarrow y = \frac{60 - x}{2}$.
3. Area $A = x \times y = x \left(\frac{60 - x}{2} \right) = \frac{60x - x^2}{2} = -\frac{1}{2}x^2 + 30x$.
4. This is a quadratic in x with $a = -\frac{1}{2}$, $b = 30$. The vertex gives the maximum.
 - Axis of symmetry: $x = -\frac{b}{2a} = -\frac{30}{2(-\frac{1}{2})} = \frac{30}{1} = 30$ m.
5. Find y :

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$$y = \frac{60 - 30}{2} = 15 \text{ m}$$

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$$\text{Maximum area: } (A_{\max}) = 30 \times 15 = 450 \text{ m}^2$$

Answer: Dimensions – 30 m along the river, 15 m away from the river.

% Example 5 – Discriminant & Nature of Roots (1 mark)

Determine the nature of the roots of $4x^2 + 12x + 9 = 0$.

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$$\Delta = 12^2 - 4(4)(9) = 144 - 144 = 0$$

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$(\Delta = 0)$!One repeated real root (double root).

$$\text{Root: } (x = \frac{-12 \pm \sqrt{0}}{2 \cdot 4} = \frac{-12}{8} = -\frac{3}{2})$$

5p ã Key Points to Remember

- **Always start by putting the quadratic into standard form** $(ax^2 + bx + c = 0)$.
- **Check for a common factor** (e.g., divide by 2) before applying any method.
- **Factorisation** works quickly when the roots are integers or simple fractions.
- **Quadratic formula** works for every quadratic; remember to simplify the radical if possible.
- **Discriminant** tells you instantly whether you will get real or complex roots – a useful checkpoint for marking.
- **Completing the square** is the bridge to the quadratic formula and to vertex form; keep the steps tidy.
- **Vertex** $((h, k))$ where $(h = -\frac{b}{2a})$ and $(k = f(h))$. Use it for graph sketching and optimisation problems.
- **Sign convention:** In the formula, $(-b)$ means “negative of b”. Don’t forget the $\pm$ sign!
- **Units & context** – always attach appropriate units (metres, seconds, dollars) in application questions.

6p ã Common Exam Questions & Tips

| Question Type | Typical Marks | Quick Tips |

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| **Factorise a quadratic** (e.g., $(x^2 + 5x + 6)$) | 1–2 | Look for product = (ac) ; test pairs quickly. |

| **Solve by quadratic formula** (with large numbers) | 3–4 | Compute discriminant first; simplify " 9B &V`ore division. |

| **State nature of roots** (given $(ax^2 + bx + c)$) | 1 | Evaluate 9B öæÇ’ b æð æVVB Fò f–æB F†R &ö÷G0. |

| **Find vertex & axis of symmetry** | 2–3 | Use $(h = -b/(2a))$; plug back to get (k) . |

| **Application problem** (max/min area, projectile) | 4–6 | Translate word problem into quadratic, then use vertex for max/min. |

| **Complete the square** (e.g., $(2x^2 + 8x + 5 = 0)$) | 2–3 | Factor out ‘a’ first if $(a \neq 1)$. |

| **Sketch the graph** (including intercepts) | 5–6 | Find roots (x intercepts), vertex, axis, and sign of ‘a’ (opens up/down). |

| **Complex roots** (9B ôâ ó ’ Â " 2 Â Write answer as $(p \pm qi)$; remember $(i^2 = -1)$. |

General exam strategies

1. **Read the question twice** – identify which method the examiner expects.
2. **Write the standard form** clearly; a neat layout earns marks even if you make a small arithmetic slip.
3. **Show one line of reasoning** – e.g., “Since 9B ôâ ó ’ Â F†R W V F–öâ † 2 Gpo distinct real roots”.
- 4.

Check your answer by substituting back (quick mental check for simple roots).

- 5. Time management** allocate ~5 /min per 2 mark quadratic question; reserve 10 12 /min for a multi step application problem.

7p ã Summary

- A quadratic equation is any equation that can be written as $(ax^2+bx+c=0)$ ($(a\neq 0)$).
- **Four main solving techniques**: factorisation, completing the square, quadratic formula, and using the discriminant to assess the nature of roots.
- The **discriminant** (b^2-4ac) — $b^2-4ac > 0$ diagnostic tool.
- The **vertex form** $(a(x-h)^2+k)$ gives the turning point; the axis of symmetry is $(x=h)$.
- **Applications** (max/min, projectile motion, area optimisation) turn word problems into quadratics; the vertex yields the optimum value.
- Practice each method with past ZIMSEC/IGCSE papers – the more you recognise patterns, the faster you'll write clean, full credit solutions.

Keep practising, stay organized, and you'll master quadratics in no time!

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